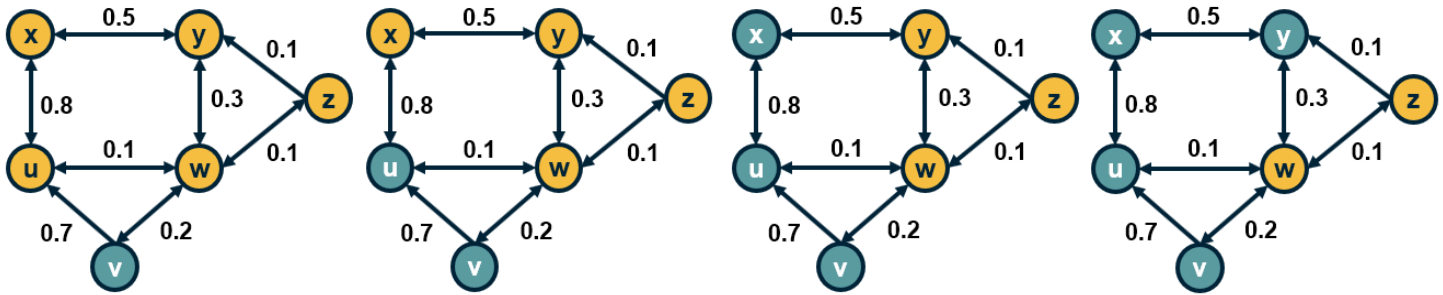


L10: Independent Contagion Model



In some cases the activation of a node v can be triggered by a single active neighbor of v , independent of the state of other neighbors of v . The Independent Contagion model is more appropriate in such problems.

Again, the state of each node can be inactive or active. If there is an edge from u to v , then the weight $w_{u,v}$ of the edge in this model represents the probability that node u activates node v when the former becomes active.

Note that node u has a single chance to activate node v . Nodes stay in the active state after they have been activated.

An important difference with the Linear Threshold model is that the Independent Contagion model is probabilistic, and so the activation cascade of a seed node v has to be described statistically.

Food For Thought

Explain why the Independent Contagion model is consistent with the Diminishing Returns theory of social influence.